

ARIAN EMAMJOMEH

Waterloo, Ontario

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EDUCATION

University of Waterloo — 3.88/4 GPA

Sep. 2025 – April 2029

Bachelors in Computer Science - President's Distinction and Paul M. Koch Scholarship Awards

Waterloo, ON

EXPERIENCE

Volvo Energy

Summer 2027

Incoming Software Engineering Intern

Gothenburg, Sweden

- Incoming SWE intern on Volvo's **BESS (Battery Energy Storage Systems)** Product Development team.

Inductive Solution

Jan. 2026 – Aug. 2026

Software Engineering Intern

Boston, MA

- Developed **Aura**, a conversational AI assistant serving consumers and businesses across web, mobile, SMS, and email.
- Built out the **Python LLM backend** that orchestrates multi-turn conversations and **35+ function-calling tools** (email, calendar, CRM, bookkeeping) over per-user encrypted SQLite storage that isolates each customer's data.
- Hardened the runtime for production with atomic message deduplication across **Redis** reconnects and failover across **4 LLM providers** with per-provider cooldowns, eliminating duplicate replies and mid-deploy conversation drops.
- Engineered an autonomous coordination engine that books property showings end-to-end across **4 contact channels** (ShowingTime, WhatsApp, SMS, email), parsing agent replies, auto-renegotiating scheduling conflicts by priority, and writing confirmed times to calendar, collapsing multi-property tours into a single request to the assistant.
- Built an LLM-driven browser agent executing real **MLS/ShowingTime** bookings through a sandboxed **Playwright** worker with human-approved, idempotent writes, cutting the live MLS read failure rate from **~33% to 0%**.

Harvard Medical School Research Study

Sep. 2024 – Sep. 2025

Research Lead

Boston, MA

- Led a cross-national study on how social media usage affects adolescent behaviour and sleep health across **Iran and Canada**, supervised by Amirreza Haghighi, Lecturer of Medicine at **Harvard Medical School**.
- Collected and analyzed survey responses from **150+ students** using the **PSQI, SMAS, and ERQ** instruments to correlate screen time, anxiety levels, and sleep quality, producing findings relevant to **7 million adolescents**.

PROJECTS

INSU - Automated Insulin Delivery Platform | Swift, Gemini API

- Engineered a **closed-loop control system ("artificial pancreas")** serving the 2.5+ million North Americans with Type 1 Diabetes, automating biological feedback loops to maintain stable glucose without manual intervention.
- Built low-level Bluetooth driver integrations to interface directly with **Omnipod insulin pumps** and **Dexcom CGMs**, enabling real-time command execution and continuous biometric data ingestion without vendor SDKs.
- Integrated **Google Gemini** to estimate carbohydrates from meal photos via computer vision with structured confidence intervals, and connected to Apple HealthKit to surface glucose, insulin, and carb history in one view.

GitHub Secret Scanner & Security Analysis Tool | Python, Gemini API, GitHub API, Entropy Analysis

- Developed a Python-based security scanner that streams GitHub commit history from GH Archives to detect **leaked credentials**, alerting **500+ developers** to live secrets exposed in their public repositories so they could be rotated.
- Built an entropy-based detection algorithm using Shannon entropy to **identify 2000+ unknown secret patterns**.
- Created an LLM-integrated false-positive filter **reducing false positives by 90%+** with the Gemini API.

NBA Predictor - ML 🧠 | Python, FastAPI, React, scikit-learn, SQLAlchemy, NBA API

- Engineered a full-stack ML application predicting NBA game outcomes with **70% accuracy** using a **Logistic Regression** model trained on **4,000+ historical games**, with fresh predictions published for every game all season.
- Built a scalable data pipeline processing **31 engineered features** per game including team statistics, recent form, head-to-head records, and home court advantage to generate real-time predictions for every fixture.
- Developed a **RESTful API with FastAPI** serving predictions through **8 endpoints**, achieving **sub-100ms latency** for model inference and feature extraction from a SQLite database, served to a React dashboard front end.

TECHNICAL SKILLS

Languages: Python, C, JavaScript, TypeScript, Racket, HTML/CSS | **Spoken:** English, French, Farsi

Frameworks/Tools: Git, React, React Native, NextJS, SwiftUI, Figma, Expo Go, FastAPI, Redis, Playwright, Supabase, Linux